

Sentiment-Augmented Quarterly Stock Prediction

A leakage-aware supervised learning pipeline for predicting 63-day sector-relative stock performance after quarterly filing events

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INEG 41403 Data Mining
Univ. of Arkansas, Spring '26

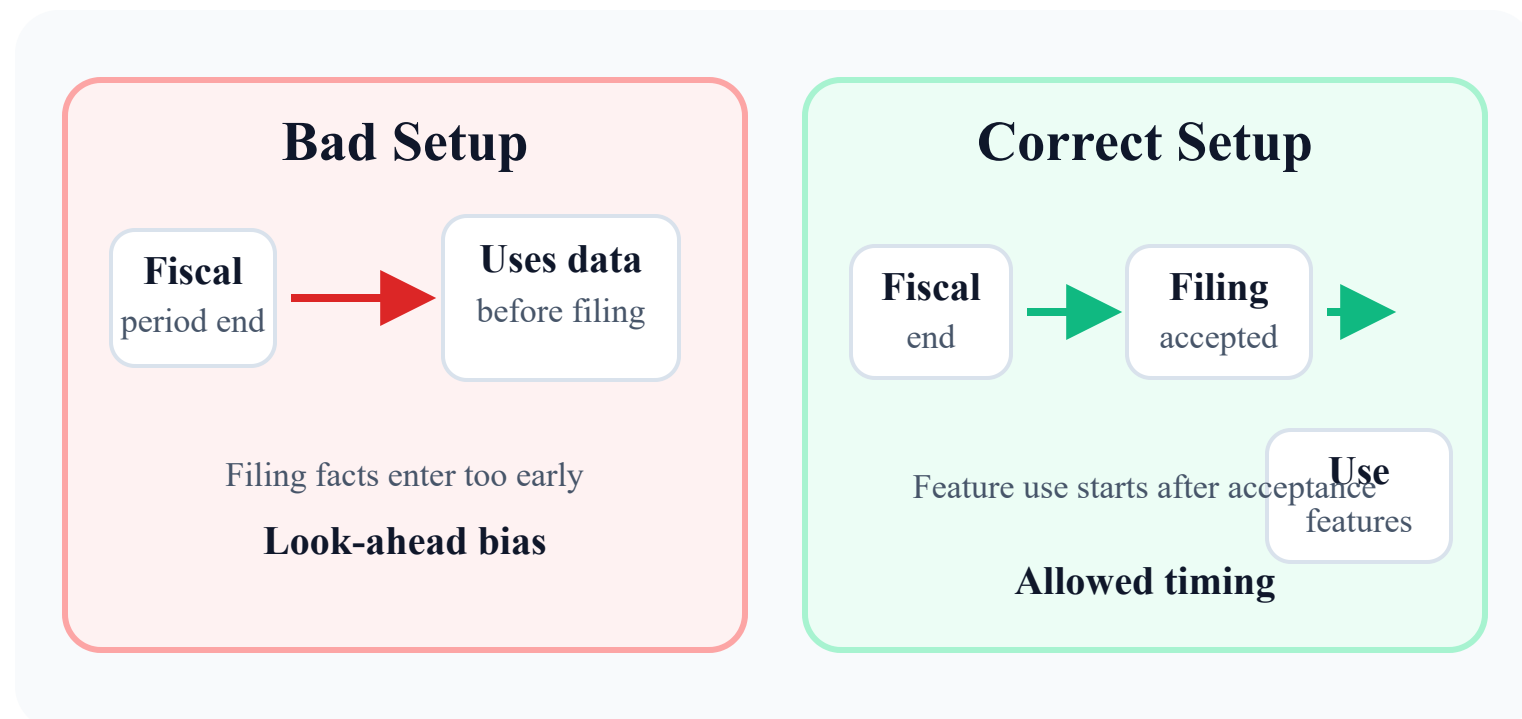
Thesis: This project tests whether accounting fundamentals, market context, and sentiment/event-specific features can predict whether a stock will outperform its sector after a quarterly filing event.

Why This Problem Is Difficult

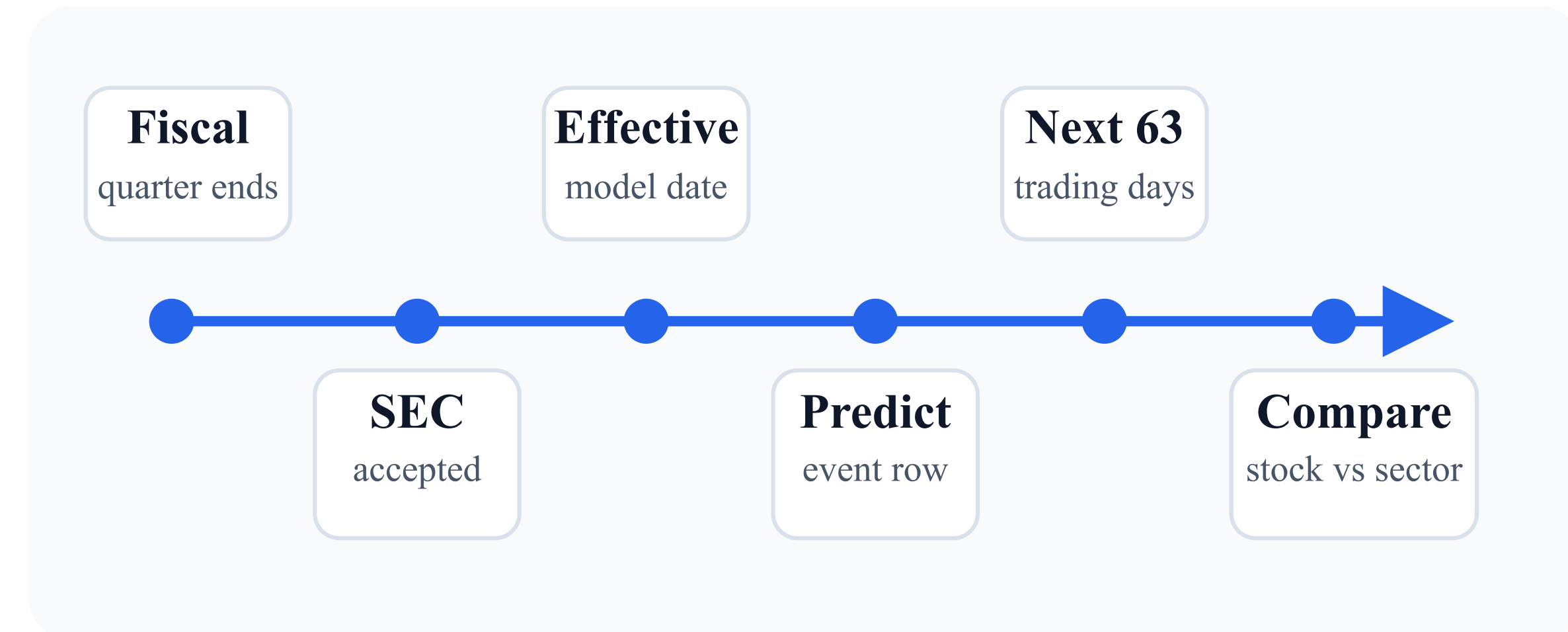
Financial prediction is noisy and vulnerable to look-ahead bias. A model can appear predictive if it accidentally uses information that would not have been available at the time.

- No random train/test splits.
- No future information in features.
- Labels must not overlap validation windows.

Leakage Control

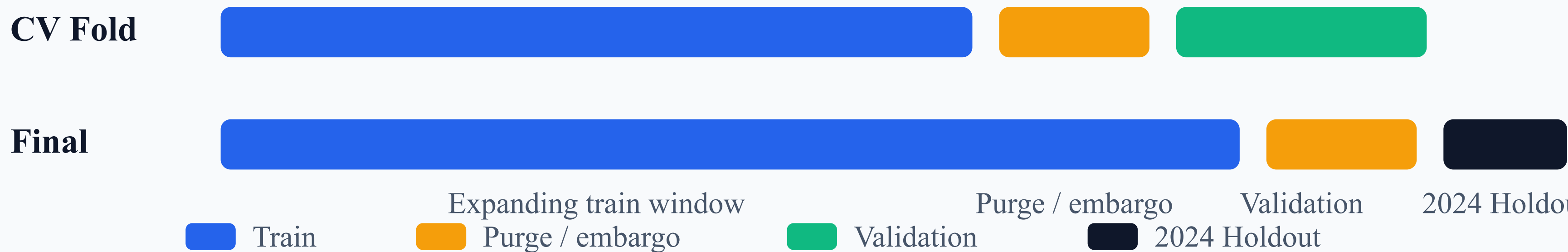


Event Timing & Validation



Features become usable only after the filing/event is tradable.

Purged Expanding-Window Validation



Purged expanding-window CV with fixed 2024 holdout.

Research Question

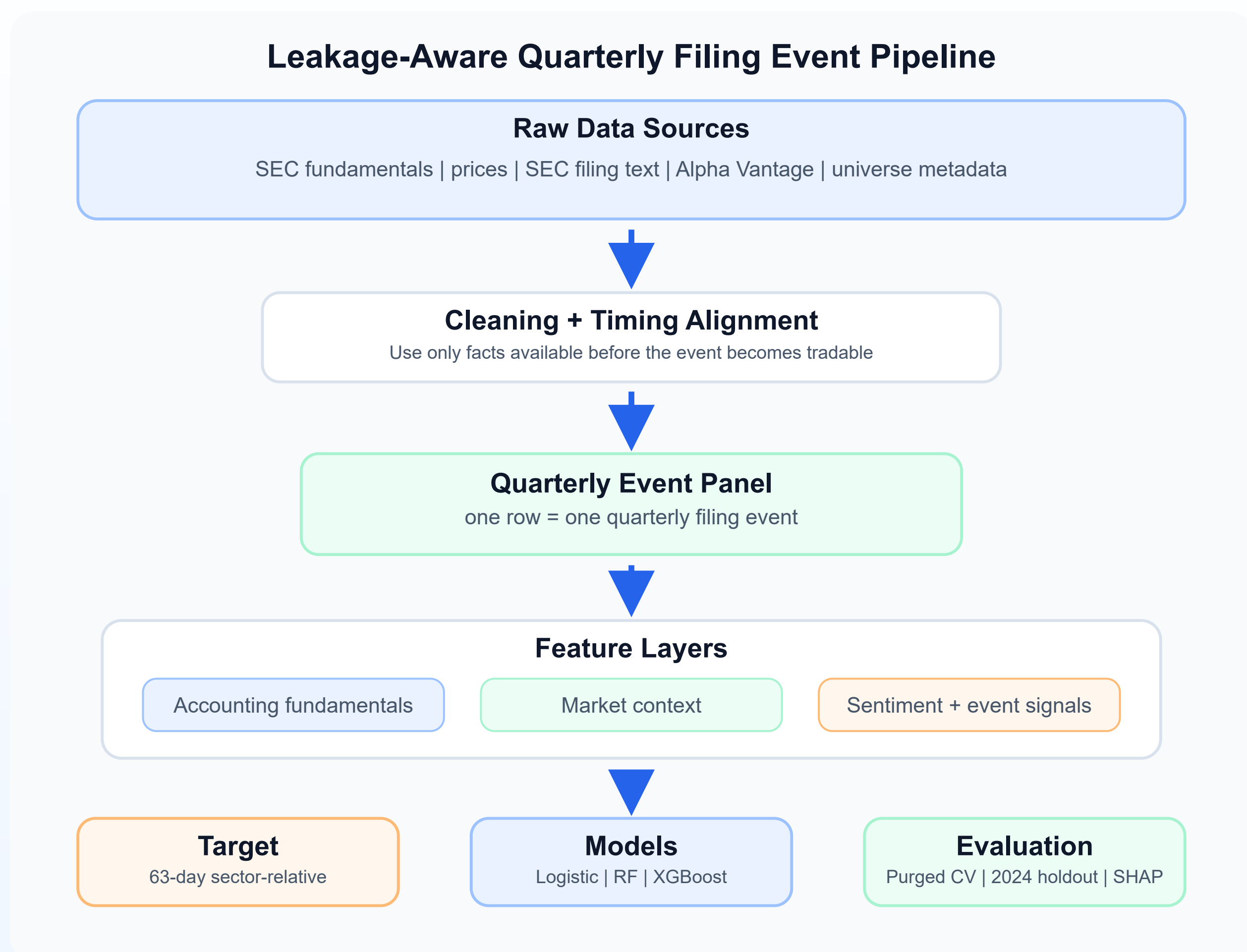
Can a machine learning model predict whether a company will outperform its sector over the next 63 trading days after a quarterly filing event?

Do sentiment and event-specific features add predictive value beyond fundamentals and market context?

Data Sources

Source	Role
SEC EDGAR	Fundamentals & text
Market prices	Returns & volatility
Alpha Vantage	EPS surprise, revisions
Capital IQ	Sector context

Modeling Pipeline



What the Experiments Showed

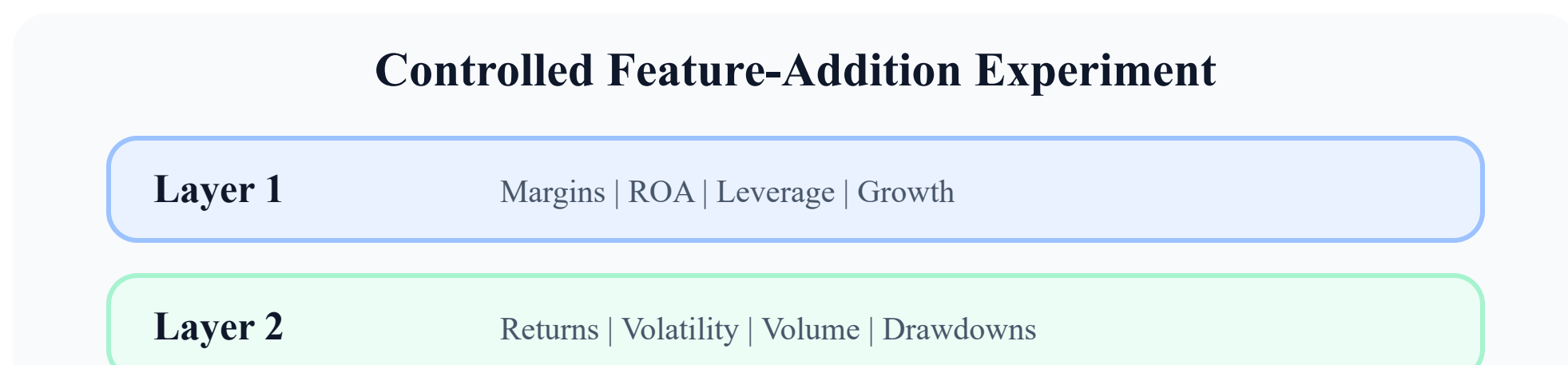


The strongest sentiment result comes from a legacy 21-day benchmark; the final contract uses the stricter 63-day sector-relative target.

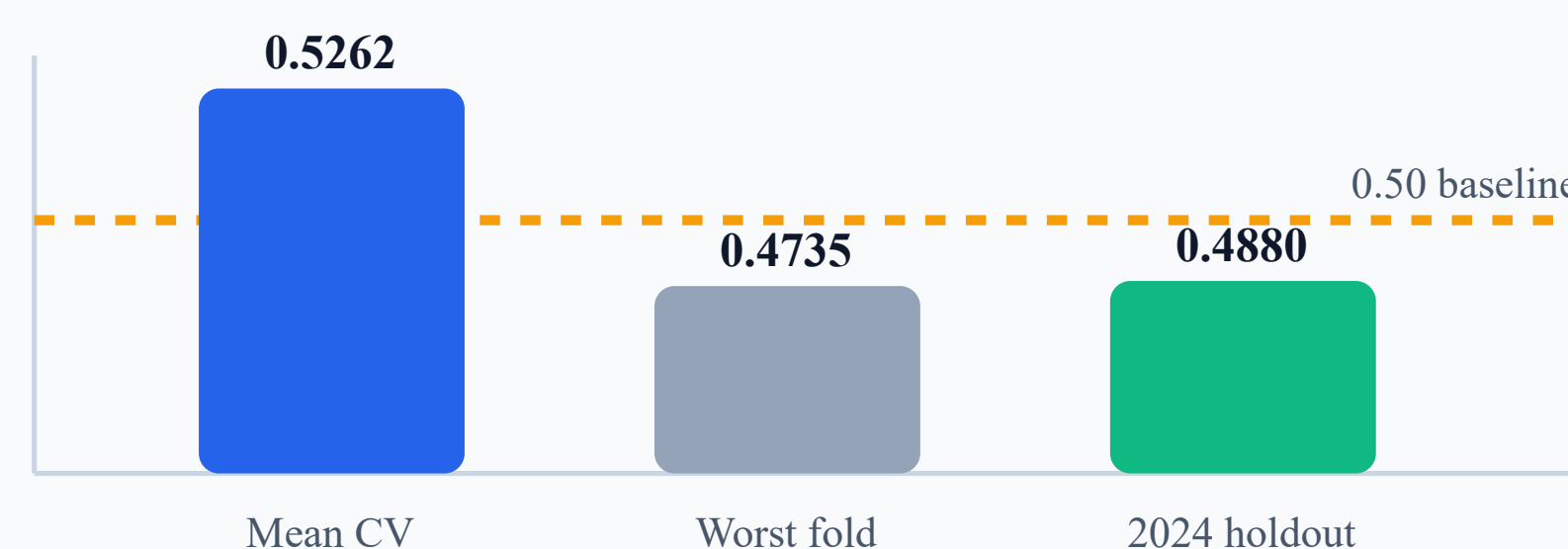
Selected Benchmark Evidence

63d Stability Candidate XGBoost CV AUC: 0.5262 Holdout: 0.4880	Legacy 21d Sentiment Random Forest Holdout AUC: 0.5367 Legacy 21d benchmark, not final 63d contract.
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Feature Design



63d Stability Candidate AUC



Sentiment Feature Comparison

